

Functions of Two Variables, Limits, Continuity, and Derivatives

Lecture 3 for Multivariable Calculus

Contents

1	Lecture Scope and Goals	2
2	Functions in Higher Dimension: Focus on $\mathbb{R}^2 \rightarrow \mathbb{R}$	2
2.1	Domain and Graph	2
3	Distance, Open Sets, and Neighborhoods in $\mathbb{R}^2$	3
3.1	Open Ball in $\mathbb{R}^2$	3
3.2	Open Set	3
4	Limits of Functions of Two Variables	3
4.1	Definition (ε - δ)	3
4.2	Equivalent Sequential Criterion	4
4.3	Worked Examples	4
5	Continuity of Two-Variable Functions	8
5.1	Quick Review: Continuity in One Variable	8
5.2	Comparison with Two Variables	9
5.3	Pointwise Definition	10
5.4	Computation Rules	10
5.5	Theorem: Continuity of Composition (Different Combinations)	10
5.6	Topological (Open Set) View	11
5.7	Examples of Continuity	11
6	Derivative in Two Variables: Best Linear Approximation	11
6.1	Partial Derivatives: First Step Toward Derivatives	11
6.2	Definition of Differentiability at a Point	14
6.3	Coordinate Form	15
6.4	Total Differential and Its Relation to Partial Derivatives	15
6.5	Tangent Plane Formula	16
6.6	Examples	16
6.7	A Useful Sufficient Condition	17
6.8	Higher-Order Partial Derivatives (Order Can Matter)	17
6.9	More Examples: How to Judge Differentiability Quickly	19
7	Mini Workflow for Students	19
8	Practice Set (In-Class or Homework)	20

1 Lecture Scope and Goals

In this lecture, we focus on functions of two variables and the transition from one-variable calculus ideas to higher dimensions. The main goals are:

- understand what a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ means geometrically and analytically,
- define and compute limits of two-variable functions,
- study continuity from computational and topological viewpoints,
- introduce the derivative as a best linear approximation in two variables,
- identify common pitfalls when handling multivariable limits.

Central Theme

In one variable, approaching a point means moving from left or right. In two variables, there are infinitely many directions and curves of approach. This is why limits, continuity, and derivatives require more careful definitions.

2 Functions in Higher Dimension: Focus on $\mathbb{R}^2 \rightarrow \mathbb{R}$

A function of two variables is a rule

$$f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto f(x, y).$$

2.1 Domain and Graph

- The **domain** D is the set of input points (x, y) where f is defined.
- The **graph** is the surface in $\mathbb{R}^3$:

$$z = f(x, y).$$

- **Level curves** are planar traces:

$$f(x, y) = c.$$

Example 1 (polynomial).

$$f(x, y) = x^2 + xy + y^2.$$

Domain: all of $\mathbb{R}^2$.

Level curves $x^2 + xy + y^2 = c$ are ellipses (for $c > 0$).

Example 2 (restricted domain).

$$f(x, y) = \sqrt{1 - x^2 - y^2}.$$

Domain: disk $x^2 + y^2 \leq 1$. The graph is the upper hemisphere of radius 1.

Example 3 (quotient).

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

Domain: $\mathbb{R}^2 \setminus \{(0, 0)\}$.

3 Distance, Open Sets, and Neighborhoods in $\mathbb{R}^2$

For points $p = (x_1, y_1)$ and $q = (x_2, y_2)$ in $\mathbb{R}^2$, define distance by

$$d(p, q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \|p - q\|.$$

3.1 Open Ball in $\mathbb{R}^2$

For center $a \in \mathbb{R}^2$ and radius $r > 0$,

$$B_r(a) = \{p \in \mathbb{R}^2 : \|p - a\| < r\}.$$

3.2 Open Set

A set $U \subseteq \mathbb{R}^2$ is **open** if for every $a \in U$, there exists $r > 0$ such that

$$B_r(a) \subseteq U.$$

Intuition: points in an open set are interior points; no boundary point is included.

Examples.

- Open disk: $x^2 + y^2 < 1$ is open.
- Closed disk: $x^2 + y^2 \leq 1$ is not open.
- Upper half-plane: $\{(x, y) : y > 0\}$ is open.

4 Limits of Functions of Two Variables

Let $f : D \rightarrow \mathbb{R}$ and let $a = (a_1, a_2)$ be a limit point of D .

4.1 Definition (ε - δ)

We say

$$\lim_{(x, y) \rightarrow (a_1, a_2)} f(x, y) = L$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever

$$0 < \|(x, y) - (a_1, a_2)\| < \delta,$$

we have

$$|f(x, y) - L| < \varepsilon.$$

4.2 Equivalent Sequential Criterion

The limit equals L if and only if for every sequence $(x_n, y_n) \rightarrow (a_1, a_2)$ with $(x_n, y_n) \neq (a_1, a_2)$,

$$f(x_n, y_n) \rightarrow L.$$

This criterion is powerful for proving limits *do not* exist.

4.3 Worked Examples

Example 4 (limit exists).

$$\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) = 0.$$

Reason: $|x^2 + y^2| = x^2 + y^2 = \|(x, y)\|^2 \rightarrow 0$.

Example 5 (path dependence: limit does not exist).

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

Along $y = 0$, $f(x, 0) = 1$. Along $x = 0$, $f(0, y) = -1$. Different path-limits imply no limit at $(0, 0)$.

Example 6 (a classical positive example).

$$f(x, y) = \frac{x^2 y}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

Claim:

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0.$$

Estimate:

$$|f(x, y)| = \frac{x^2 |y|}{x^2 + y^2} \leq |y| \leq \sqrt{x^2 + y^2} = \|(x, y)\| \rightarrow 0.$$

Example 7 (testing many lines is not enough).

$$f(x, y) = \frac{x^2 y}{x^4 + y^2}, \quad (x, y) \neq (0, 0).$$

For line $y = mx$,

$$f(x, mx) = \frac{mx^3}{x^4 + m^2 x^2} = \frac{mx}{x^2 + m^2} \rightarrow 0.$$

But for curve $y = x^2$,

$$f(x, x^2) = \frac{x^4}{x^4 + x^4} = \frac{1}{2}.$$

So the limit does not exist.

Example 8 (full limit exists, but one iterated limit does not). Define

$$f(x, y) = \begin{cases} x \sin\left(\frac{1}{y}\right), & y \neq 0, \\ 0, & y = 0. \end{cases}$$

At $(0, 0)$,

$$|f(x, y)| \leq |x| \rightarrow 0,$$

so

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0.$$

However,

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right)$$

does not exist, because for each fixed $x \neq 0$, the inner limit

$$\lim_{y \rightarrow 0} x \sin\left(\frac{1}{y}\right)$$

does not exist.

Example 9 (full limit exists, both iterated limits fail). Define

$$f(x, y) = \begin{cases} x \sin\left(\frac{1}{y}\right) + y \sin\left(\frac{1}{x}\right), & xy \neq 0, \\ 0, & xy = 0. \end{cases}$$

Then

$$|f(x, y)| \leq |x| + |y| \rightarrow 0,$$

so

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0.$$

But for fixed $x \neq 0$, the term $x \sin(1/y)$ makes $\lim_{y \rightarrow 0} f(x, y)$ fail; similarly for fixed $y \neq 0$, the term $y \sin(1/x)$ makes $\lim_{x \rightarrow 0} f(x, y)$ fail. Therefore neither iterated limit exists.

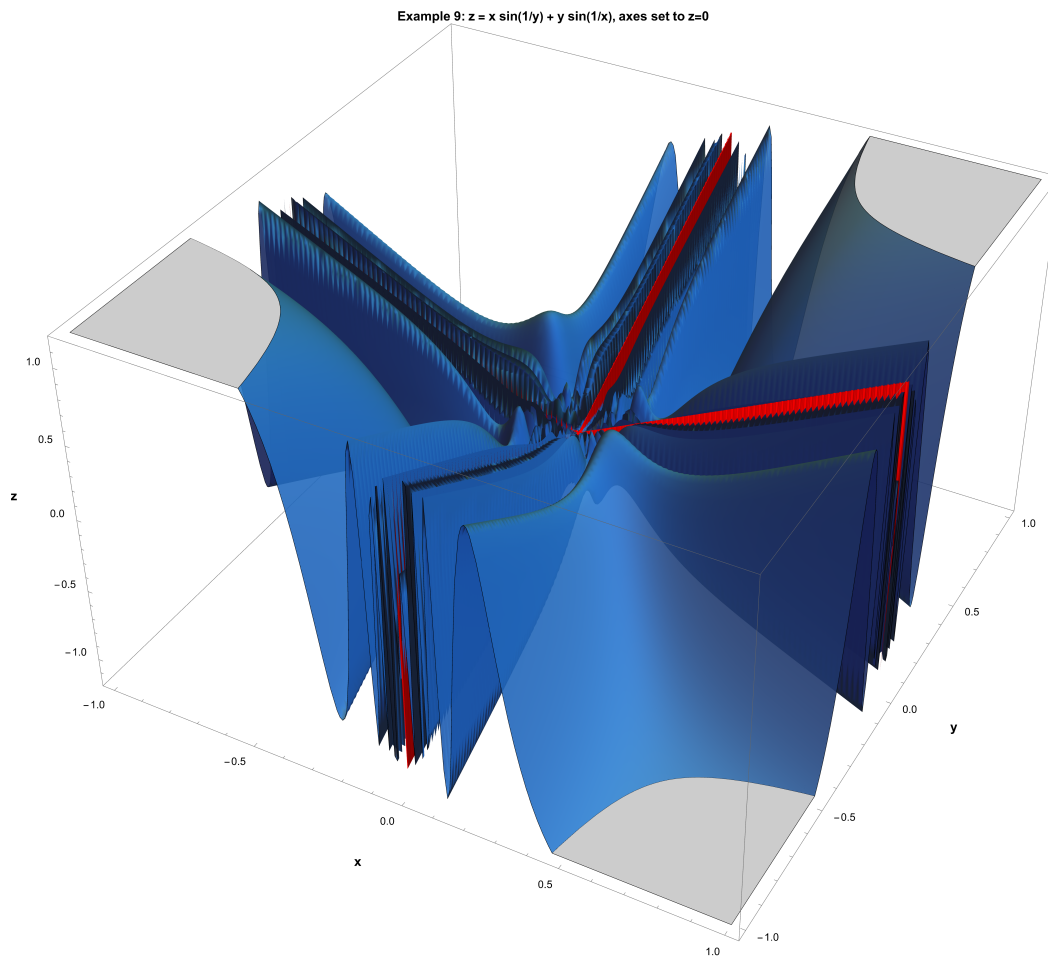


Figure 1: 3D visualization for Example 9: $f(x, y) = x \sin(1/y) + y \sin(1/x)$ (with $f = 0$ on the axes). The red lines mark $x = 0$ and $y = 0$.

Example 10 (jump discontinuity along a line). Consider

$$g(x, y) = \begin{cases} x + y, & y > x, \\ x + y + 2, & y < x, \\ x + y + 1, & y = x. \end{cases}$$

This is a two-variable function with a jump across the oblique line $y = x$. At any point (a, a) on that line,

$$\lim_{(x,y) \rightarrow (a,a), y > x} g(x, y) = 2a, \quad \lim_{(x,y) \rightarrow (a,a), y < x} g(x, y) = 2a + 2,$$

so the limit does not exist on that line.

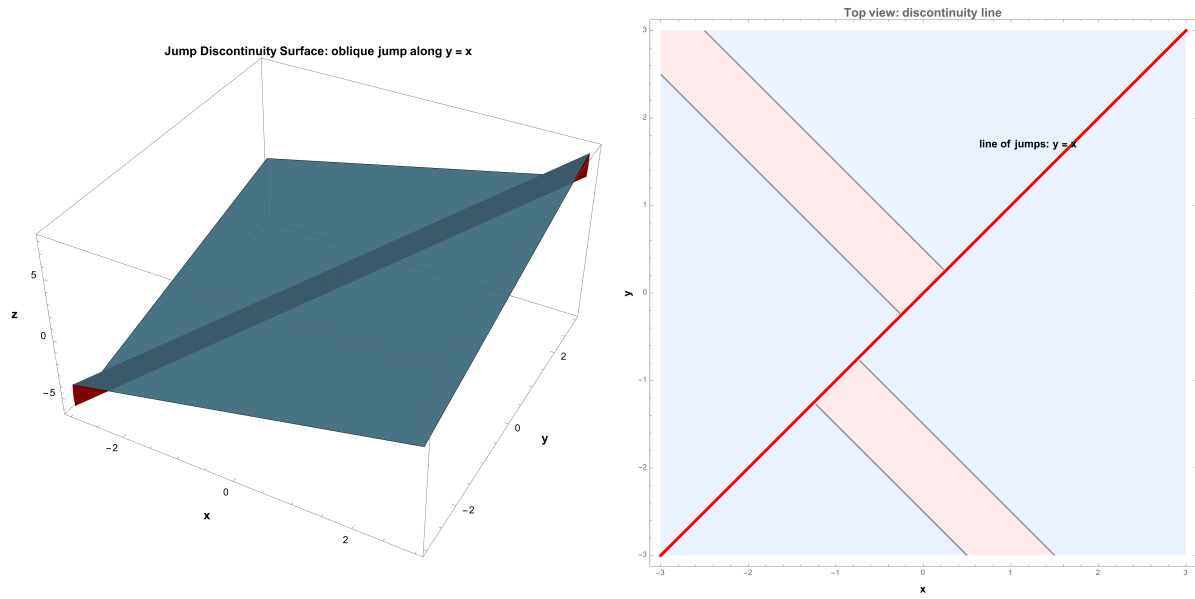


Figure 2: Nontrivial jump-discontinuity example. Left: two offset planes over regions $y > x$ and $y < x$. Right: top view with the red jump line $y = x$.

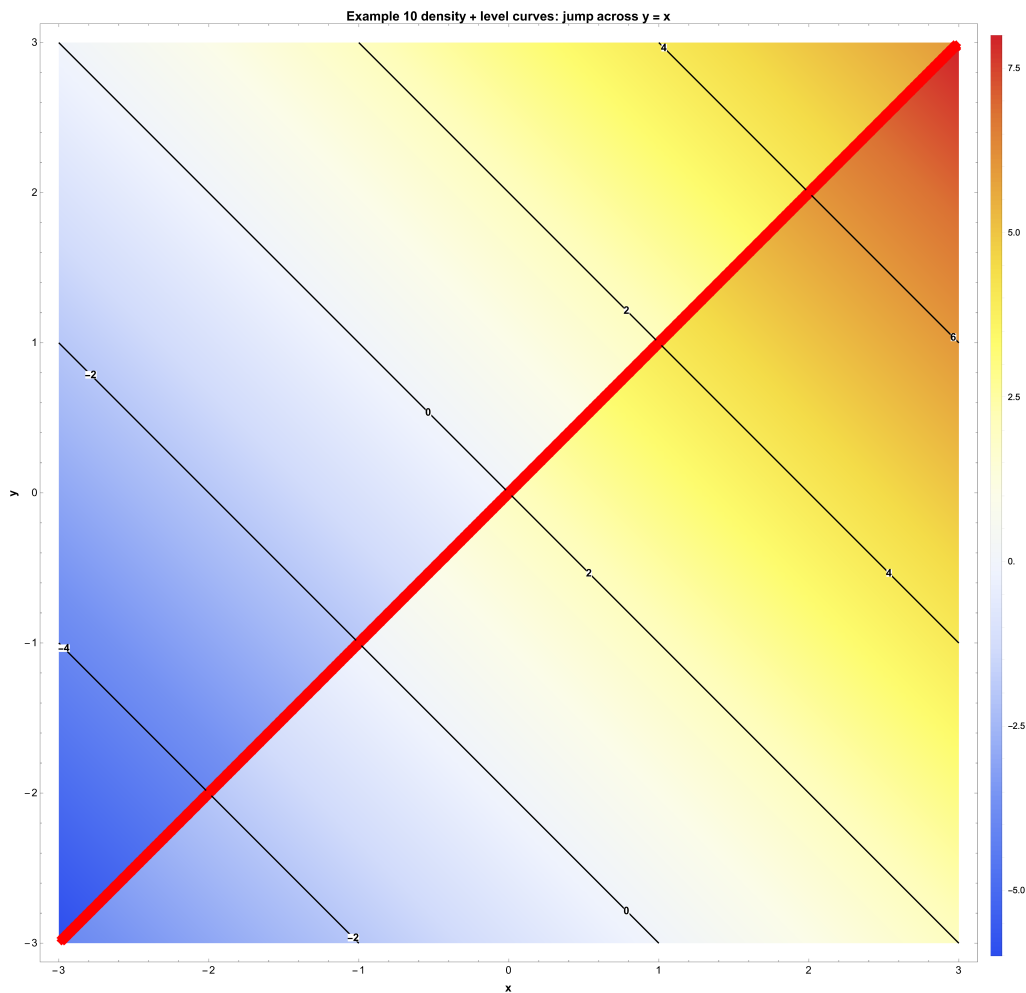


Figure 3: Density (heatmap) and level curves for Example 10. On each side of $y = x$, level sets are lines; crossing $y = x$ causes a jump by 2, so matching level curves are offset across the red jump line.

Pitfall 1

Checking only paths of the form $y = mx$ can miss nonlinear approaches. If all lines give the same value, the limit still may fail to exist.

Pitfall 2

Do not substitute $(0, 0)$ directly into a formula with denominator 0 and conclude anything about the limit. Undefined value at the point and existence of limit are different questions.

5 Continuity of Two-Variable Functions

5.1 Quick Review: Continuity in One Variable

Before moving fully to two variables, recall the one-variable picture. Let $h : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$.

- **Continuity at x_0 :** $\lim_{x \rightarrow x_0} h(x) = h(x_0)$.
- **Intermediate Value Theorem (IVT):** if h is continuous on $[a, b]$, then every value between $h(a)$ and $h(b)$ is attained.
- **Extreme Value Theorem (max/min):** if h is continuous on $[a, b]$, then h attains both a maximum and a minimum.
- **Boundedness theorem:** if h is continuous on $[a, b]$, then h is bounded on $[a, b]$.

The key reason these hold is that $[a, b]$ is **compact** (closed and bounded in $\mathbb{R}$), and also **connected**.

5.2 Comparison with Two Variables

Now let $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. The pointwise definition is analogous. In calculus, for global conclusions (boundedness and max/min), we mainly work on **compact domains**.

(A) Intermediate-value behavior. In one variable, IVT is phrased on an interval $[a, b]$. In two variables, a practical classroom version is:

If f is continuous on a region and you move continuously from point P to point Q inside that region, then f takes every value between $f(P)$ and $f(Q)$ along that motion.

So we still keep the same intuition: continuous change cannot skip intermediate values.

Example (IVT failure on a disconnected domain). Let

$$D = \{(x, y) \in \mathbb{R}^2 : x < 0\} \cup \{(x, y) \in \mathbb{R}^2 : x > 0\}$$

and define

$$f(x, y) = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

The function is continuous on D . Choose $P = (-1, 0)$ and $Q = (1, 0)$. Then

$$f(P) = -1, \quad f(Q) = 1.$$

But there is no point in D where $f = 0$, so an intermediate value is missing. This shows IVT can fail when the domain is not connected.

(B) Max/min and boundedness. For $\mathbb{R}^2$, the direct analogue of $[a, b]$ is a **compact set** $K \subseteq \mathbb{R}^2$ (closed and bounded), and this is the main domain type we care about in calculus applications.

If $f : K \rightarrow \mathbb{R}$ is continuous and K is compact, then f is bounded on K and attains both global maximum and global minimum on K .

(C) Why compactness matters (pitfall). If the domain is not compact, max/min can fail even for continuous functions.

$$f(x, y) = x^2 + y^2 \quad \text{on } \mathbb{R}^2$$

is continuous and has a minimum 0 at $(0, 0)$, but no maximum.

$$f(x, y) = x \quad \text{on } \{(x, y) : x^2 + y^2 < 1\}$$

is bounded by $-1 < f < 1$, but has no attained maximum/minimum because boundary points are excluded.

5.3 Pointwise Definition

A function $f : D \rightarrow \mathbb{R}$ is continuous at $a \in D$ if

$$\lim_{(x,y) \rightarrow a} f(x,y) = f(a).$$

So continuity means: *the limit exists and equals the function value.*

5.4 Computation Rules

If f, g are continuous at a , then so are:

- $f + g$, fg , and cf ,
- $\frac{f}{g}$ when $g(a) \neq 0$,
- compositions $\phi \circ f$ when ϕ is one-variable continuous.

Hence polynomials are continuous everywhere; rational functions are continuous on domains where denominator is nonzero.

5.5 Theorem: Continuity of Composition (Different Combinations)

Composition Continuity Theorem

If F is continuous at a and G is continuous at $F(a)$, then $G \circ F$ is continuous at a .

Quick Proof (Topological Definition)

Let $F : X \rightarrow Y$ and $G : Y \rightarrow Z$ be continuous. To prove $G \circ F : X \rightarrow Z$ is continuous, take any open set $U \subseteq Z$. Since G is continuous, $G^{-1}(U)$ is open in Y . Since F is continuous, $F^{-1}(G^{-1}(U))$ is open in X . But

$$(G \circ F)^{-1}(U) = F^{-1}(G^{-1}(U)).$$

So the preimage of every open set in Z under $G \circ F$ is open in X . Therefore $G \circ F$ is continuous.

This one statement includes the main calculus combinations:

- **1D $\rightarrow$ 1D $\rightarrow$ 1D:**

$$F : \mathbb{R} \rightarrow \mathbb{R}, \quad G : \mathbb{R} \rightarrow \mathbb{R}$$

continuous $\Rightarrow G(F(x))$ continuous.

- **2D $\rightarrow$ 1D $\rightarrow$ 1D:**

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad G : \mathbb{R} \rightarrow \mathbb{R}$$

continuous $\Rightarrow G(F(x, y))$ continuous.

- **1D $\rightarrow$ 2D $\rightarrow$ 1D:**

$$F : \mathbb{R} \rightarrow \mathbb{R}^2, \quad F(t) = (u(t), v(t)), \quad G : \mathbb{R}^2 \rightarrow \mathbb{R}$$

continuous $\Rightarrow G(u(t), v(t))$ continuous in t .

- **2D $\rightarrow$ 2D $\rightarrow$ 1D:**

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad F(x, y) = (u(x, y), v(x, y)), \quad G : \mathbb{R}^2 \rightarrow \mathbb{R}$$

continuous $\Rightarrow G(u(x, y), v(x, y))$ continuous.

Quick examples.

- $F(x, y) = x^2 + y^2$, $G(s) = e^s \Rightarrow e^{x^2+y^2}$ is continuous.
- $F(t) = (\cos t, \sin t)$, $G(x, y) = xy \Rightarrow G(F(t)) = \cos t \sin t$ is continuous.
- $F(x, y) = (x + y, xy)$, $G(u, v) = \frac{u}{1 + v^2} \Rightarrow \frac{x + y}{1 + x^2 y^2}$ is continuous.

5.6 Topological (Open Set) View

A function $f : D \rightarrow \mathbb{R}$ is continuous if and only if for every open set $V \subseteq \mathbb{R}$, the preimage

$$f^{-1}(V) = \{(x, y) \in D : f(x, y) \in V\}$$

is open in D (with the subspace topology).

Why this matters. This abstract definition works in all metric spaces and is independent of coordinates.

5.7 Examples of Continuity

Example 11.

$$f(x, y) = e^{x^2+y^2} \sin(xy)$$

is continuous on $\mathbb{R}^2$ because it is built from continuous operations.

Example 12 (piecewise). Define

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

By Example 6, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$, so f is continuous at the origin.

6 Derivative in Two Variables: Best Linear Approximation

6.1 Partial Derivatives: First Step Toward Derivatives

For a function $f(x, y)$, we can vary one variable at a time.

Definition at (a, b) .

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}, \quad f_y(a, b) = \lim_{k \rightarrow 0} \frac{f(a, b+k) - f(a, b)}{k},$$

when these limits exist.

- f_x is the rate of change in the x -direction (hold $y = b$ fixed).
- f_y is the rate of change in the y -direction (hold $x = a$ fixed).

Geometrically, $f_x(a, b)$ and $f_y(a, b)$ are slopes of trace curves on planes $y = b$ and $x = a$.

Geometry picture (simple example). Take

$$f(x, y) = x^2 + y^2,$$

and the point $(1, 1, 2)$ on the surface $z = f(x, y)$.

- Fix $y = 1$: trace curve is $z = x^2 + 1$, so slope at $x = 1$ is

$$f_x(1, 1) = 2.$$

- Fix $x = 1$: trace curve is $z = 1 + y^2$, so slope at $y = 1$ is

$$f_y(1, 1) = 2.$$

So each partial derivative is the tangent-line slope of a different planar slice through the same surface point.

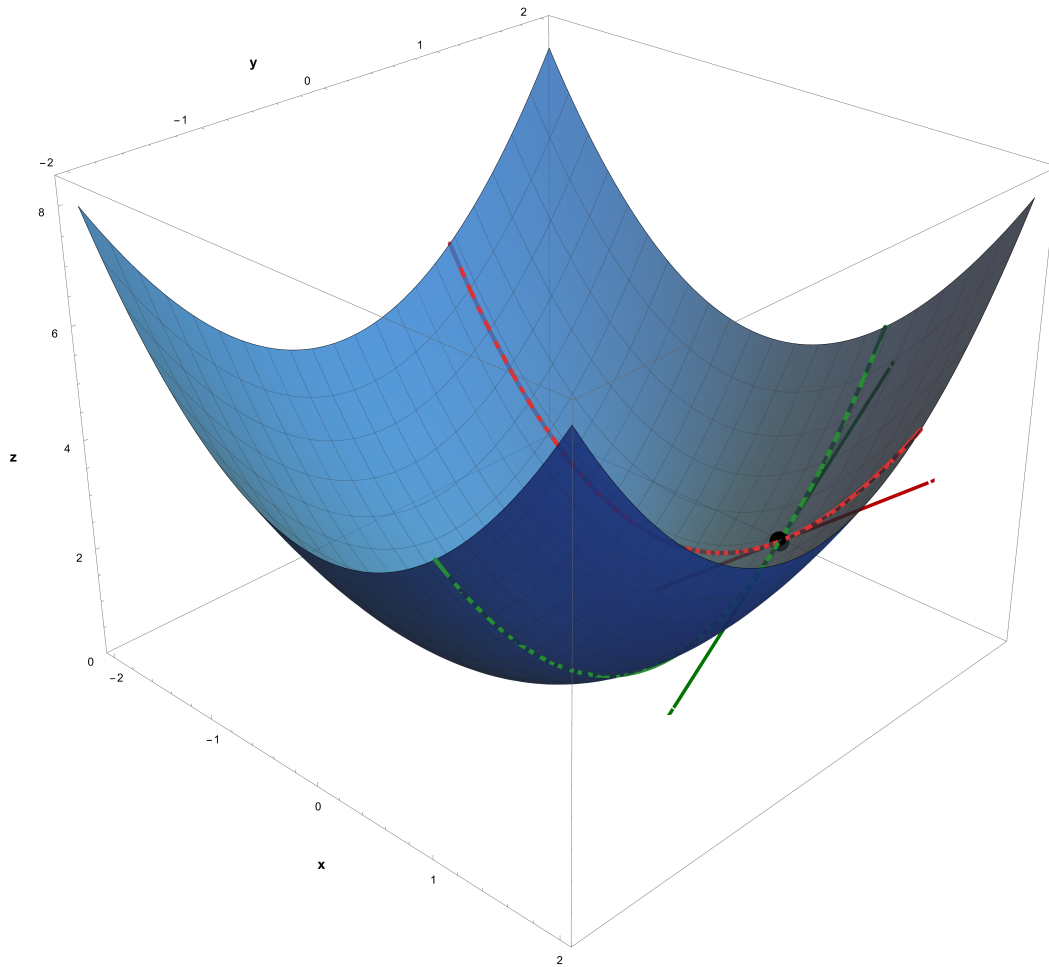


Figure 4: Geometric meaning of partial derivatives for $z = x^2 + y^2$ at $(1, 1, 2)$. Solid red/green curves are traces on $y = 1$ and $x = 1$; dashed lines are tangent lines giving slopes $f_x(1, 1)$ and $f_y(1, 1)$.

Notation.

$$f_x = \frac{\partial f}{\partial x}, \quad f_y = \frac{\partial f}{\partial y}.$$

If f is differentiable, the total differential is

$$df = f_x dx + f_y dy.$$

Examples.

- If $f(x, y) = x^2y + \sin(xy)$, then

$$f_x = 2xy + y \cos(xy), \quad f_y = x^2 + x \cos(xy).$$

- If $f(x, y) = |x| + y^2$, then at $(0, 0)$:

$$f_y(0, 0) = 0 \quad \text{but} \quad f_x(0, 0) \text{ does not exist.}$$

So even existence of one partial derivative does not imply existence of the other.

6.2 Definition of Differentiability at a Point

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $a = (a_1, a_2)$. We say f is differentiable at a if there exists a linear map

$$L : \mathbb{R}^2 \rightarrow \mathbb{R}$$

such that

$$f(a + h) - f(a) - L(h) = o(\|h\|) \quad \text{as } h \rightarrow 0.$$

Equivalently,

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a) - L(h)}{\|h\|} = 0.$$

The map L is the derivative $Df(a)$.

Theorem (Common Sufficient Condition)

If f_x and f_y exist in a neighborhood of (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .

Proof (short). Let $h = (h_1, h_2)$ and write

$$\Delta f := f(a + h_1, b + h_2) - f(a, b).$$

Split the increment:

$$\Delta f = [f(a + h_1, b + h_2) - f(a, b + h_2)] + [f(a, b + h_2) - f(a, b)].$$

Apply one-variable mean value theorem to each bracket:

$$f(a + h_1, b + h_2) - f(a, b + h_2) = f_x(a + \theta_1 h_1, b + h_2) h_1,$$

$$f(a, b + h_2) - f(a, b) = f_y(a, b + \theta_2 h_2) h_2,$$

for some $\theta_1, \theta_2 \in (0, 1)$. Hence

$$\Delta f = f_x(a, b) h_1 + f_y(a, b) h_2 + R(h),$$

where

$$R(h) = (f_x(a + \theta_1 h_1, b + h_2) - f_x(a, b)) h_1 + (f_y(a, b + \theta_2 h_2) - f_y(a, b)) h_2.$$

Because f_x, f_y are continuous at (a, b) , the two bracketed differences go to 0 as $h \rightarrow 0$. Therefore

$$\frac{|R(h)|}{\|h\|} \leq |f_x(a + \theta_1 h_1, b + h_2) - f_x(a, b)| \frac{|h_1|}{\|h\|} + |f_y(a, b + \theta_2 h_2) - f_y(a, b)| \frac{|h_2|}{\|h\|} \rightarrow 0.$$

So $\Delta f = f_x(a, b) h_1 + f_y(a, b) h_2 + o(\|h\|)$, which is exactly differentiability at (a, b) .

Theorem (Zero Partial Derivatives Imply Constant on a Region)

Let $D \subseteq \mathbb{R}^2$ be a region (open and connected), and let $f : D \rightarrow \mathbb{R}$ be differentiable on D . If

$$f_x(x, y) = 0, \quad f_y(x, y) = 0 \quad \text{for all } (x, y) \in D,$$

then f is constant on D .

Proof (short). Take any two points $P, Q \in D$. Since D is a region, we can choose a smooth curve

$$\gamma : [0, 1] \rightarrow D, \quad \gamma(0) = P, \quad \gamma(1) = Q.$$

Define $g(t) = f(\gamma(t))$. By the one-variable chain rule,

$$g'(t) = \nabla f(\gamma(t)) \cdot \gamma'(t) = f_x(\gamma(t)) x'(t) + f_y(\gamma(t)) y'(t) = 0.$$

Hence g is constant on $[0, 1]$, so

$$f(P) = g(0) = g(1) = f(Q).$$

Because P, Q are arbitrary, f is constant on D .

6.3 Coordinate Form

In two variables, if differentiable,

$$Df(a)(h_1, h_2) = f_x(a)h_1 + f_y(a)h_2,$$

so

$$f(a + h) \approx f(a) + f_x(a)h_1 + f_y(a)h_2.$$

The gradient is

$$\nabla f(a) = (f_x(a), f_y(a)),$$

and

$$Df(a)(h) = \nabla f(a) \cdot h.$$

6.4 Total Differential and Its Relation to Partial Derivatives

Let $f = f(x, y)$ be differentiable at (a, b) . For small increments

$$\Delta x = x - a, \quad \Delta y = y - b,$$

the function increment satisfies

$$\Delta f := f(a + \Delta x, b + \Delta y) - f(a, b) = f_x(a, b) \Delta x + f_y(a, b) \Delta y + o\left(\sqrt{(\Delta x)^2 + (\Delta y)^2}\right).$$

The linear main part

$$df = f_x(a, b) dx + f_y(a, b) dy$$

is called the **total differential**.

Key relation.

- Partial derivatives are the coefficients of the total differential.
- Geometrically, df is the first-order change of f along a small move (dx, dy) .
- Equivalently, $df = \nabla f(a, b) \cdot (dx, dy)$.

Example. Let

$$f(x, y) = x^2y + \sin(xy).$$

Then

$$f_x = 2xy + y \cos(xy), \quad f_y = x^2 + x \cos(xy),$$

so at $(1, 2)$,

$$f_x(1, 2) = 4 + 2 \cos 2, \quad f_y(1, 2) = 1 + \cos 2.$$

Hence the total differential at $(1, 2)$ is

$$df = (4 + 2 \cos 2) dx + (1 + \cos 2) dy.$$

Therefore, for small changes $(\Delta x, \Delta y)$ near $(1, 2)$,

$$\Delta f \approx (4 + 2 \cos 2) \Delta x + (1 + \cos 2) \Delta y.$$

6.5 Tangent Plane FormulaFor $z = f(x, y)$ at $(a, b, f(a, b))$, the tangent plane is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

In-Class Problem

Find the tangent plane of

$$z = x^2 + y^2$$

at the point $(1, 2, 5)$.**6.6 Examples****Example 13 (differentiable polynomial).**

$$f(x, y) = x^2 + 3xy - y^2.$$

At $(1, 2)$,

$$f_x = 2x + 3y, \quad f_y = 3x - 2y,$$

so

$$\nabla f(1, 2) = (8, -1), \quad Df(1, 2)(h_1, h_2) = 8h_1 - h_2.$$

Local linearization:

$$f(1 + h_1, 2 + h_2) \approx f(1, 2) + 8h_1 - h_2.$$

Example 14 (partials exist but not differentiable at origin). Define

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Both $f_x(0, 0)$ and $f_y(0, 0)$ exist and equal 0, but

$$f(t, t) = \frac{t^2}{\sqrt{2t^2}} = \frac{|t|}{\sqrt{2}},$$

which is not $o(\sqrt{t^2 + t^2}) = o(|t|)$. Hence f is not differentiable at $(0, 0)$.

Pitfall 3

Existence of partial derivatives at a point does *not* guarantee differentiability at that point.

6.7 A Useful Sufficient Condition

If f_x and f_y exist in a neighborhood of a and are continuous at a , then f is differentiable at a .

6.8 Higher-Order Partial Derivatives (Order Can Matter)

Once first partial derivatives exist, we can differentiate again.

Second-order partials.

$$f_{xx} = \frac{\partial}{\partial x}(f_x), \quad f_{yy} = \frac{\partial}{\partial y}(f_y), \quad f_{xy} = \frac{\partial}{\partial y}(f_x), \quad f_{yx} = \frac{\partial}{\partial x}(f_y).$$

Here f_{xy} and f_{yx} are **mixed partial derivatives**.

Important Theorem (Clairaut/Schwarz)

If f_{xy} and f_{yx} are continuous in a neighborhood of a point, then

$$f_{xy} = f_{yx}$$

at that point.

Short proof. Fix a point (a, b) and define, for small h, k ,

$$\Phi(h, k) := f(a + h, b + k) - f(a + h, b) - f(a, b + k) + f(a, b).$$

By applying one-variable fundamental theorem of calculus twice,

$$\Phi(h, k) = \int_0^h \int_0^k f_{xy}(a + u, b + v) \, dv \, du,$$

and also

$$\Phi(h, k) = \int_0^k \int_0^h f_{yx}(a + u, b + v) \, du \, dv.$$

Divide by hk :

$$\frac{\Phi(h, k)}{hk} = \frac{1}{hk} \int_0^h \int_0^k f_{xy}(a + u, b + v) \, dv \, du = \frac{1}{hk} \int_0^k \int_0^h f_{yx}(a + u, b + v) \, du \, dv.$$

Let $(h, k) \rightarrow (0, 0)$. Continuity of f_{xy} and f_{yx} gives

$$\lim_{(h,k) \rightarrow (0,0)} \frac{\Phi(h, k)}{hk} = f_{xy}(a, b) = f_{yx}(a, b).$$

Hence $f_{xy}(a, b) = f_{yx}(a, b)$.

Second short proof (using Mean Value Theorem). Define

$$g(y) := f(a + h, y) - f(a, y).$$

Then

$$\frac{\Phi(h, k)}{k} = \frac{g(b + k) - g(b)}{k}.$$

By the one-variable mean value theorem in y , for some η between b and $b + k$,

$$\frac{\Phi(h, k)}{k} = g'(\eta) = f_y(a + h, \eta) - f_y(a, \eta).$$

Apply the mean value theorem again in x to the function $x \mapsto f_y(x, \eta)$: for some ξ between a and $a + h$,

$$\frac{\Phi(h, k)}{hk} = f_{yx}(\xi, \eta).$$

Similarly, exchanging the roles of x and y , one gets

$$\frac{\Phi(h, k)}{hk} = f_{xy}(\tilde{\xi}, \tilde{\eta}),$$

for points $(\tilde{\xi}, \tilde{\eta})$ in the same small rectangle. Let $(h, k) \rightarrow (0, 0)$. By continuity of f_{xy} and f_{yx} ,

$$f_{yx}(a, b) = \lim_{(h,k) \rightarrow (0,0)} \frac{\Phi(h, k)}{hk} = f_{xy}(a, b).$$

Hence $f_{xy}(a, b) = f_{yx}(a, b)$.

So under sufficient smoothness, the order of differentiation does *not* matter. But without those conditions, the order can matter.

Example where order matters at one point. Define

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

At $(0, 0)$, one can compute

$$f_{xy}(0, 0) = -1, \quad f_{yx}(0, 0) = 1.$$

Hence

$$f_{xy}(0, 0) \neq f_{yx}(0, 0),$$

so the differentiation order matters here.

6.9 More Examples: How to Judge Differentiability Quickly

Example A (smooth polynomial: differentiable everywhere).

$$f(x, y) = x^3 + 2xy - y^2.$$

All partial derivatives are polynomials, hence continuous everywhere. Therefore f is differentiable at every point of $\mathbb{R}^2$.

Example B (continuous but not differentiable at one point).

$$f(x, y) = \sqrt{x^2 + y^2}.$$

At $(0, 0)$, the function is continuous. But

$$\frac{f(th) - f(0, 0)}{|t|} = \frac{|t| \|h\|}{|t|} = \|h\|$$

does not come from a linear map in h , so f is not differentiable at $(0, 0)$.

Example C (differentiable despite piecewise definition).

$$f(x, y) = \begin{cases} x^2 + y^2, & x^2 + y^2 \leq 1, \\ 2x^2 + 2y^2 - 1, & x^2 + y^2 > 1. \end{cases}$$

On the circle $x^2 + y^2 = 1$, both formulas give value 1, but gradients are different:

$$\nabla(x^2 + y^2) = (2x, 2y), \quad \nabla(2x^2 + 2y^2 - 1) = (4x, 4y).$$

So f is continuous on $\mathbb{R}^2$, but not differentiable on the boundary circle.

Example D (key positive criterion in practice).

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Near $(0, 0)$,

$$0 \leq |f(x, y)| = \frac{x^2 y^2}{x^2 + y^2} \leq \frac{(x^2 + y^2)^2}{4(x^2 + y^2)} = \frac{x^2 + y^2}{4}.$$

This estimate gives a quadratic error bound, which is strong evidence toward differentiability at the origin (with derivative 0).

7 Mini Workflow for Students

When facing a new two-variable problem, follow this checklist:

1. Identify domain and possible singular points.
2. For limits: try estimates with $\|(x, y)\|$, then test strategic paths.
3. For continuity: compare limit and function value at the target point.
4. For derivatives: compute partials, then check differentiability (not only existence of partials).
5. Interpret results geometrically (level curves, tangent plane, local linear model).

8 Practice Set (In-Class or Homework)

1. Determine whether each limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}, \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2}.$$

2. Study continuity at $(0,0)$ for

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

3. For $f(x, y) = \ln(1 + x^2 + y^2)$, compute $\nabla f(1, -1)$ and write tangent plane at $(1, -1, f(1, -1))$.
4. Give an example where all directional derivatives at $(0,0)$ exist but the function is not differentiable at $(0,0)$.

Takeaway

For two-variable calculus, the key upgrade from one variable is geometric thinking with neighborhoods in the plane. Limits and continuity depend on *all* nearby directions, and derivatives mean *linear approximation*, not just partial derivative formulas.